



Research Paper

A fractional-order multiple-model type-2 fuzzy control for interconnected power systems incorporating renewable energies and demand response

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ABSTRACT

Frequency regulation in Multi-region Interconnected Power Systems (MIPS), incorporating wind turbine systems, energy storage units, and demand response, is a challenging control problem. The problem involves maintaining grid stability, integrating variable renewable energy sources, enhancing grid resilience, optimizing energy storage and demand response capabilities, and ensuring regulatory compliance. Effective frequency regulation is a key problem for reliable and sustainable power system operation. In this paper, complex and uncertain dynamics in various components of a MIPS are modeled using multiple first-order dynamic fractional-order Type-2 Fuzzy Logic Systems (T2-FLSs). The models are evaluated, and the best possible dynamic model is chosen. With the best possible model, an optimal controller is designed. The stability and optimality analysis are presented through a Linear Matrix Inequality (LMI) approach. The adaptive laws of T2-FLSs are derived such that some LMI-based conditions are satisfied. The designed controller does not depend on the dynamics of MIPS. The uncertainties of time-varying load, wind energy, and solar power are modeled using T2-FLSs. The suggested LMI technique presents adaptation laws for T2-FLSs to ensure stability in the presence of natural disturbances and errors. The designed scheme is validated by applying to a practical 39-bus IEEE test system that includes the wind farms, demand response, and energy storage systems. The simulation results under various conditions verify the usefulness of the suggested controller. The comparisons with conventional controllers demonstrate that the suggested approach is more effective under uncertainties and natural perturbations.

1. Introduction

A multi-region interconnected power system comprises control areas that represent a set of generators that operate at the same frequency. These generators control both frequency regulation and power difference of all interconnected regions that are exchanging power by contracts. In some cases, the primary frequency controller is inadequate for frequency regulation, especially in a MIPS. Thus, it is necessary to consider the frequency control function to regulate the generators' output power and keep the power up that is exchanged among interconnected regions. A classical Automatic Generation Control (AGC) includes a Proportional Integrator (PI) and a Proportional Integrator Derivative (PID) (Peng et al., 2023). The deviation control function is used to enhance the transient response after frequency event, while

the integrator controller turns the frequency changes back to zero. The disadvantage of conventional controllers is that their parameters become steady over time, making the system less resistant to performance constraints.

Recently, researchers have paid considerable attention to modern intelligent approaches since they can cover a broader range of operations, and increase performance and resilience of the system through online optimization than traditional controllers (Modabbernia et al., 2022; Xue et al., 2024). Another way to control automatic generation is controller design H_∞ , and multi-objective control H_2/H_∞ (Begum et al., 2024). The H_∞ is applied to the uncertain system for improving the stability, while H_2 is used to reduce area control error and gain rapid frequency response with low overshoot. However, since AGCs,

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based on H_∞ and H_2/H_∞ , need high-order dynamic control systems, they are not feasible for industrial purposes. On the other hand, modern control requires knowing all the state variables in the system. Therefore, it is not convenient for large-scale interconnected power systems that feature many state variables. In addition, in adaptive and robust control methods, there is a need to know the exact and complete model of the system. For this reason, they lack efficiency for real systems. Fluctuations in renewable energy sources pose significant challenges to power system performance.

An increase in renewable energy generation demands more frequency response services. Meeting the demands of frequency response depending on common structure necessitates constructing conventional and costly power systems and limits the capability of the plants to adapt to renewable productions. The load-side response is a cost-efficient method that can determine frequency response duties in a scheme that is like frequency-sensitive plants. It can be applicable in decreasing total electricity power generation throughout the highest energy demand hours, presently generated by high-priced power systems like gas turbines (Pinson et al., 2014). Load response is an ecologically safe mechanism since it can replace production units at peak demand hours and thus reduce carbon dioxide emissions (Aunedi et al., 2013). While more conventional power plants are being replaced by renewable generators, an energy storage system should be considered. With the help of an energy storage system, renewable generators can be reliable for increasing power quality and energy management.

Nowadays, most energy storage systems have been combined with large-scale plants to enhance their Alternating Current (AC) output (Hansen and Hansen, 2007). For example, storage systems outfitted with the battery are used in wind farms to reduce wind turbine generator output changes that can suitably stabilize outputs with short-term oscillators (López-Queija et al., 2022; Xia et al., 2023). In addition, acceptable storage systems have several features, such as quick access to energy, high energy capacity, long-lasting, and low price. On the other hand, the technologies available in the energy storage system areas hardly put together all these acceptable features simultaneously. Since there is uncertainty in real power systems like modeling errors, load changes, errors that result from linearization, and lack of access to the system information, one of the main challenges that control engineering researchers encounter is Nonlinear Systems (NS) due to uncertain dynamics.

Control NS without uncertain dynamics has been extensively studied. For example in Yang et al. (2020), the back-stepping design and control design were developed by using Lyapunov functions. In Huang et al. (2022), the control sliding method was evaluated for robotic and tracking use (Jafari Fesharaki et al., 2020; Liu et al., 2020). In Yin et al. (2016), the neural network was used for uncertainty estimation. In addition, with the help of fuzzy systems of stimulus failures and the nonlinearity of the dead band, the adaptive controller was considered as uncertainty in Jing and Yang (2019). However, in the proposed methods, the neural network system's plainness and the use of the type-1 FLS were not provided to deal with uncertainties, and they were considered to be a special form of indeterminate dynamics. In recent years, many researchers have paid attention to type-2 FLS including control systems (Zhao et al., 2020, 2019), modeling of physical systems (Zarandi et al., 2019) image processing (Martínez et al., 2019), and predicting issues (Chen and Wang, 2019). More attention has only been paid to certain uncertainties and perturbations while ignoring un-modeled dynamics.

The literature review shows that:

- The integration of fuzzy control specifically for interconnected power systems with a high penetration of renewable energy sources and demand response programs has not been comprehensively studied.
- Development and optimization of fuzzy control algorithms to effectively manage the variability and uncertainty associated with renewable energy generation and demand response actions in interconnected power systems require more investigation.
- The impact of fuzzy control on system stability, reliability, and efficiency in the context of interconnected power systems with a significant share of renewable energy resources has not been studied well.
- There are not many studies on adaptive control systems that can resist natural disturbances of renewable energy sources and time-varying demand.

From the review above, in the current study, a T2-FLS based controller tuned by a LMI technique is suggested over various robust control approaches due to its robustness, ability to handle non-linearity, computational efficiency, superior performance, and adaptability to expert knowledge. The main contributions of this work are as follows.

- A new multi-model control system is proposed which increases the identification accuracy.
- First-order T2-FLS dynamic models are proposed for the online identification of unspecified dynamics.
- The fractional-order modeling and controlling systems are designed.
- The multi-region interconnected power systems are studied, and the stability is analyzed using a LMI technique.

The rest of the paper is organized as follows. In Section 2, the energy storage systems are described. The demand side response is analyzed in Section 3. The suggested T3-FLS is explained in Section 4. The general description of power dynamics is presented in Section 5. The main control problem is detailed in Section 6. The stability is analyzed in Section 7. The simulations and conclusions are given in Sections 8 and 9, respectively.

2. Energy storage system

Energy Storage Systems (ESSs), as a beneficial solution for decreasing the frequency deviation, are considered to be affected by renewable generators. In addition, ESSs can reduce the load level, provide rotating storage, and even can run after a complete shutdown (Ummels et al., 2008). Hydraulic pumps, batteries, magnetic superconductors, compressed air, and flywheels are the main types of available ESSs (Świerczyński et al., 2010). Since Battery ESSs (BESSs) are the most extensive type of storage system, they are the proposed storage system for the response of load frequency in our study. A BESS is a type of electrochemical storage system. It is charged by the chemical energy to supply the electrical energy. In addition, even in an urban area, there are no geographical restrictions on installing a battery storage system. Therefore, BESSs can be considered as the most suitable storage systems with rapid response and high-performance efficiency. Typical BESSs consist of power converter systems that connect Direct Current (DC) voltage battery banks to AC three-phase voltage systems. At times when the system is in excessive frequency or low-frequency conditions, a dynamic model of the battery storage system consumes energy from renewable generators for the frequency response purveyance.

3. Demand side response

The Demand Side Response (DSR) plays a significant role in the power system if it is employed appropriately. As described in Babahajiani et al. (2016), the demand side response can be described as the way the final consumers change their pattern of energy consumption in response to the signal. Controllable loads can be considered as another solution that can lead to balancing supply and demand. They can provide various load management methods consisting of load shifting, strategic load increasing, peak clipping, valley filling and so on Hojati and Gazor (2002). Supply-side management reduces the margins of the generation and enhances the efficiency of the power transmission and distribution network. Based on the sudden power outage, load-side

response can properly apply a primary/secondary frequency response to the system, which can be attained by increasing or decreasing the power usage of some flexible units, such as electric cars, heat pumps, ventilators, and other types of systems (Er and Chin, 2000; Wang et al., 2002).

4. Type-2 fuzzy theory

T2-FLSs are the generalized type-1 systems, where the Membership Functions (MFs) of forward and backward sections are of the second type (Cho et al., 2002). The output value of the type-2 MF can take a value between 0 and 1 that allows the membership degree itself to be a fuzzy number. This extra degree of freedom can lead to a better modeling of uncertainties. Implementing T2 rules and systems can be challenging due to the increased complexity involved. One of the main challenges in T2-FLSs is handling the extra uncertainty from the second dimension of the MF. This requires sophisticated algorithms to work with higher-dimensional MFs. The usual methods used for type-1 MFs need to be adapted to handle this added complexity. Another complexity aspect in T2-FLSs is optimizing the system parameters. Tuning the MFs, rule bases, and inference mechanisms is crucial for achieving optimal performance. The computational demands of T2-FLSs can also be a challenge in real-time applications. Efficient algorithms and data structures are essential to ensure quick processing of inputs and outputs. Balancing accuracy and speed may be necessary, as more complex computations can take longer to execute. In the present paper, un-repetitive learning rules are suggested for training of T2-FLSs.

The Footprint of Uncertainty (FOU) in the open mode can be defined with the terms upper/lower membership functions (see Fig. 1). Using the second type of MFs in the antecedent/consequent parts has some advantages compared to the first type. The output of type-2 MFs is fuzzy and can be modeled with more uncertainty and more complexity. A type-2 MF with a fuzzy secondary membership degree is preferred over several type-1 MFs because type-2 MFs reduce the number of rules, optimize control performance, and prepare better control signals. The type-2 fuzzy sets are represented mathematically as Eqs. (1) and (2), where v and u are the primary and secondary memberships, respectively:

$$\tilde{A} = \left\{ (v, u), \zeta_{\tilde{A}}(v, u) : \forall v \in V, \forall u \in J_v \subseteq [0, 1] \right\}, \quad (1)$$

where $0 \leq \zeta_{\tilde{A}}(v, u) \leq 1$. $\tilde{A}$ can also be written as (Mendel and John, 2002):

$$\tilde{A} = \int_{v \in V} \int_{u \in J_v \subseteq [0, 1]} \zeta_{\tilde{A}}(v, u) / (v, u). \quad (2)$$

The FOU is the union of all primary memberships (Mendel and John, 2002):

$$\text{FOU}(\tilde{A}) = \bigcup_{v \in V} J_v = \{(v, u) : u \in J_v \subseteq [0, 1]\}, \quad (3)$$

where

$$J_v = \left\{ (v, u) : u \in [\underline{\mu}_{\tilde{A}}(v), \bar{\mu}_{\tilde{A}}(v)] \right\}. \quad (4)$$

The FOU can also be written as

$$\text{FOU}(\tilde{A}) = \bigcup_{v \in V} [\underline{\mu}_{\tilde{A}}(v), \bar{\mu}_{\tilde{A}}(v)]. \quad (5)$$

An embedded type-2 set $\tilde{A}_e$ in $\tilde{A}$ is (Mendel and John, 2002):

$$\tilde{A}_e = \int_{v \in V} [1/u] / v \, dv, \quad u \in J_v \quad (6)$$

Let F^n be the firing intervals of the rules given by

$$F^n(v') = [\underline{f}^n(v'), \bar{f}^n(v')] = [\underline{f}^n, \bar{f}^n], \quad n \in \{1, 2, \dots, N\}, \quad (7)$$

where

$$\underline{f}^n(v') = \underline{\mu}_{\tilde{V}_1^n}(V'_1) * \dots * \underline{\mu}_{\tilde{V}_1^n}(V'_1), \quad (8)$$

$$\bar{f}^n(v') = \bar{\mu}_{\tilde{V}_1^n}(V'_1) * \dots * \bar{\mu}_{\tilde{V}_1^n}(V'_1). \quad (9)$$

The weighted firing interval is defined as

$$W_{\cos}(v') = \bigcup_{\substack{f^n \in F^n(v') \\ u^n \in W^n}} \frac{\sum_{n=1}^N w^n f^n}{\sum_{n=1}^N w^n} = [w_l, w_r], \quad (10)$$

where

$$w_l = \frac{\sum_{n=1}^L w^n \bar{f}^n + \sum_{n=L+1}^N w^n \underline{f}^n}{\sum_{n=1}^L \bar{f}^n + \sum_{n=L+1}^N \underline{f}^n} \quad (11)$$

$$w_r = \frac{\sum_{n=1}^R \bar{w}^n \underline{f}^n + \sum_{n=R+1}^N \bar{w}^n \bar{f}^n}{\sum_{n=1}^R \underline{f}^n + \sum_{n=R+1}^N \bar{f}^n} \quad (12)$$

and “cos” denotes the center-of-sets (Mendel, 2007; Debnath et al., 2018).

5. Power system dynamics

Interconnected power systems involve multiple areas linked through high-voltage transmission lines (and sometimes direct current). Each control area is perceived as an integrated system, where each generator adapts to changes in load or generation (see Fig. 2).

Modeling wind power has two stages. The first generation scenario is based on the Monte Carlo scenario and then the reduction scenario. In the first stage, the average value of wind power is predictive, and the uniform relation of the Gaussian function is used as a probability distribution function.

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}, \quad (13)$$

where μ and σ are the mean and standard deviation, respectively. The backward method is used for the reduction scenario in the second stage. In this stage, with the obtained generation scenarios and their probabilities from the first stage, the scenarios with low probabilities are omitted to select important scenarios. Assume that

$$d_{m,\hat{m}} = \min\{1, \|m - \bar{m}\|, \|m - \bar{m}\| \|m - \hat{m}\|\} \quad (14)$$

where $m \in \{1, 2, \dots, k\}$ is the number of initial scenarios in which ρ_m is the function of the probability of each scenario, $d_{m,\hat{m}}$ is the distance between two scenarios and $\bar{m}$ is the average value of the scenario. The suggested diagram is given in Fig. 3, where M is the set of initial scenarios and N is the set of omitted scenarios. According to the graph, the repetition of Steps 2–4 will continue until the number of reduction scenarios R is gained with probability $\hat{\rho}_q$. Then, we have the final model after the reduction scenario: $\min f(x) + \sum_{q=1}^R \hat{\rho}_q g(y_q)$.

Remark 1. The dynamics of all components are considered to be unknown and are modeled as described in the next section (see Fig. 4).

Remark 2. One of the challenges of T2-FLSs is updating the rule parameters. To decrease the computational cost, in the suggested algorithm, the iterative learning algorithm has been replaced with a stable learning algorithm. In the suggested learning scheme there is no iterative algorithm. Instead, the fuzzy rules are updated in each sample time, just one time. However, the learning rules are extracted from stability analysis.

6. Problem statement

The system output is expressed by several type-2 dynamic fuzzy models with different fractional orders, and the most effective estimation capability is selected for the control system design in every period. On the other hand, fractional calculus proves more efficient than the correct order in identifying the system dynamics. A higher-order

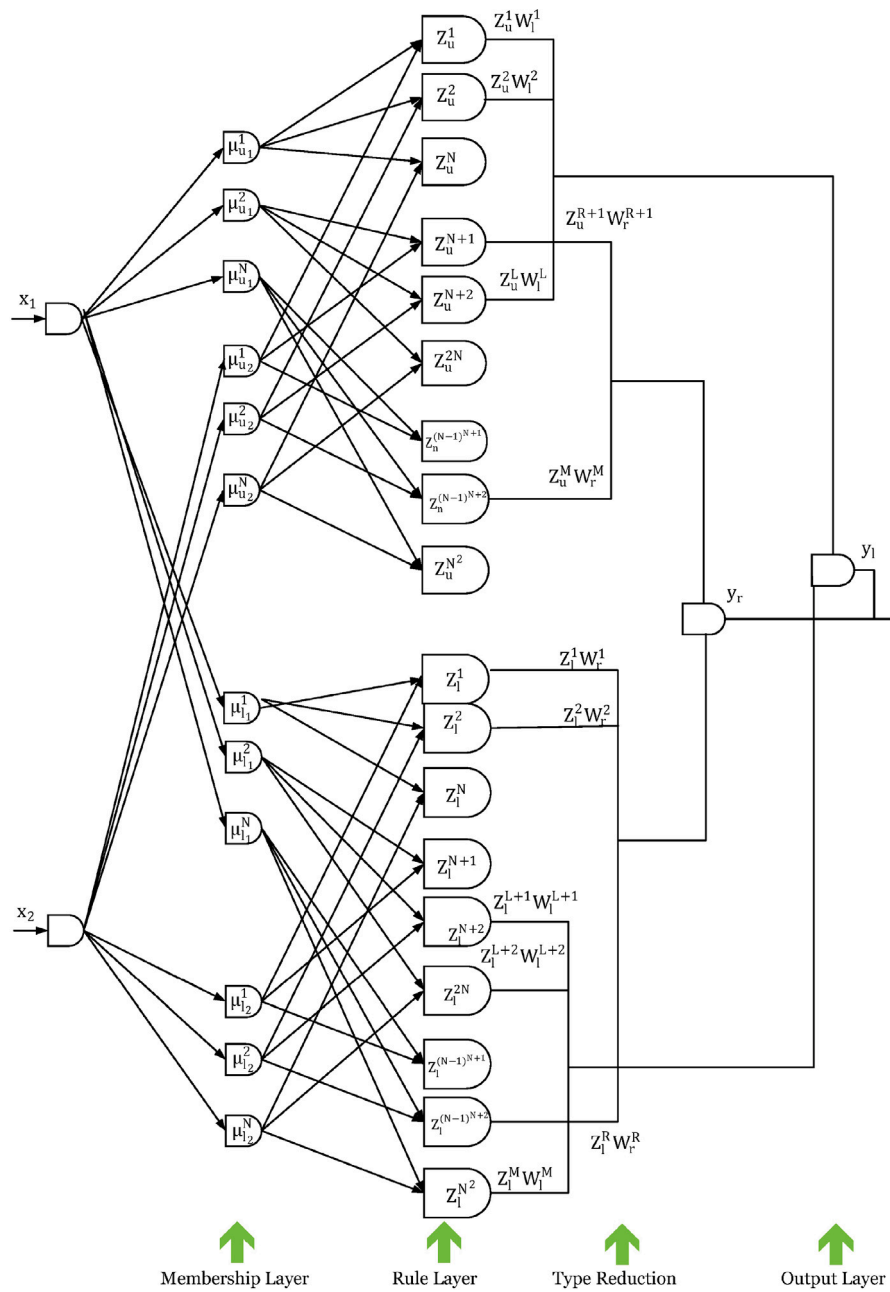


Fig. 1. The fuzzy type-2 structure.

conversion function is required to calculate a simple fractional order integral by a correct order, which increases the size and complexity of the computation. Therefore, in our proposed method, the type-1 model based on fractional-order calculus is used instead of using a high-order model to identify fractional-order calculus. Several first-order dynamic models with different fraction orders are assumed so that the value of the fraction order is not stable. Control signals are generated for the i th subsystem, and the best one will be used for the subsystem. For generating the most effective control signal, the integral error of each model should be calculated to gain the best dynamic model (See Fig. 4). Herein, it is assumed that un-modeled subsystem modes are limited. Consider the following class uncertain and un-modeled dynamics:

$$\begin{aligned} \dot{v} &= f(v_1, \dots, v_m, \theta_1, \dots, \theta_q, u) \\ \dot{\phi} &= g(v_1, \dots, v_m, \theta_1, \dots, \theta_q) \\ w &= h(v_1, \dots, v_m, \theta_1, \dots, \theta_q, u), \end{aligned} \quad (15)$$

where $v = [v_1, v_2, \dots, v_m]^T$, $\theta = [\theta_1, \theta_2, \dots, \theta_q]^T$ are the states vectors of the un-modeled dynamics, $w = [w_1, w_2, \dots, w_n]^T$ and $u = [u_1, u_2, \dots, u_n]^T$ denote the output and control vectors. f , g and h are the unknown functions.

The multiple-model for estimating the dynamics of each output is in the following form:

$$\begin{cases} C_t^{p_{11}} \hat{w}_{11}(t) = \hat{F}_{11} [w(t) | \phi_{11}(t)] - \tau_{11} \hat{e}_{11} + u_{11}(t) \\ \vdots \\ C_t^{p_{1K}} \hat{w}_{1K}(t) = \hat{F}_{1K} [w(t) | \phi_{1K}(t)] - \tau_{1K} \hat{e}_{1K} + u_{1K}(t) \\ \vdots \\ C_t^{p_{n1}} \hat{w}_{n1}(t) = \hat{F}_{n1} [w(t) | \phi_{n1}(t)] - \tau_{n1} \hat{e}_{n1} + u_{n1}(t) \\ \vdots \\ C_t^{p_{nK}} \hat{w}_{nK}(t) = \hat{F}_{nK} [w(t) | \phi_{nK}(t)] - \tau_{nK} \hat{e}_{nK} + u_{nK}(t) \end{cases} \quad (16)$$

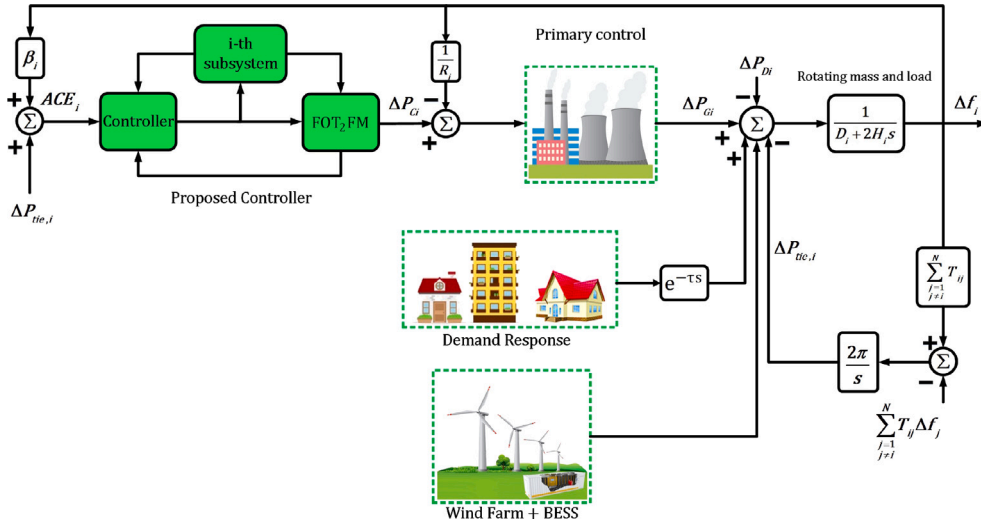
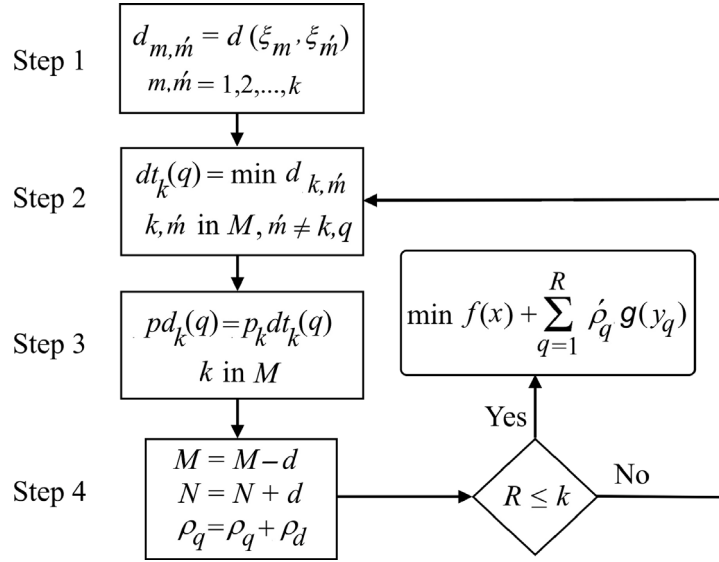
Fig. 2. The power system model for the i th area.

Fig. 3. A general view on the suggested algorithm.

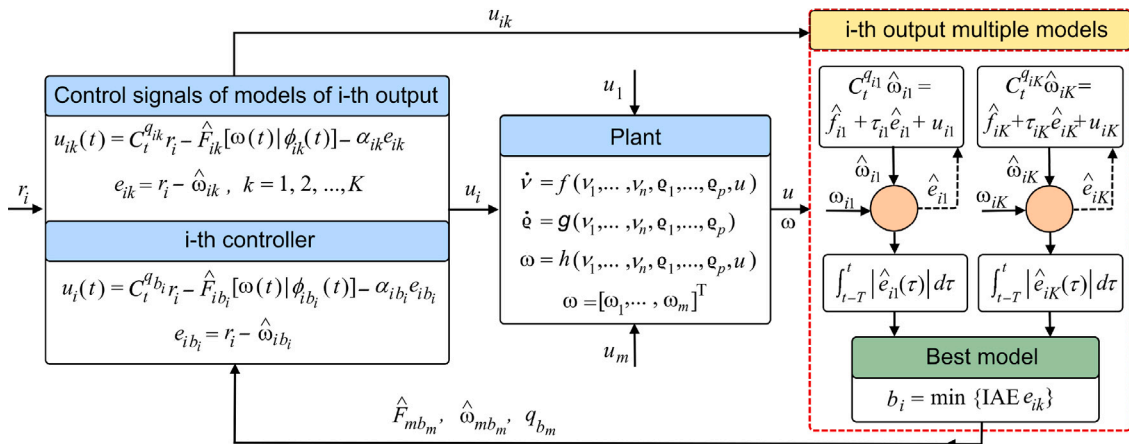


Fig. 4. The proposed structure.

where ϕ_{ik} ($i \in \{1, \dots, n\}$, $k \in \{1, \dots, K\}$) represents the parameters vector of the FLS in the k th model for the i th subsystem, τ_{ik} is defined via the stability analysis and $\hat{e}_{ik}$ is described as $\hat{e}_{ik} = \hat{w}_{ik} - w_i$ and the term $C_t^{p_i} \hat{w}_i(t)$ is the fractional-order derivative with the following form:

$$C_t^{p_i} \hat{w}_i(t) = \frac{1}{\Gamma(1-p_i)} \int_0^T \frac{\hat{w}_i^{(1)}(\zeta)}{(t-\zeta)^{p_i}} d\zeta \quad (17)$$

From system (16), the control signals can be written as:

$$u_{ik}(t) = C_t^{p_{ik}} r_i - \hat{F}_{ik}[w(t)|\phi_{ik}(t)] - \mu_{ik} e_{ik} \quad (18)$$

where u_{ik} ($i \in \{1, \dots, n\}$, $k \in \{1, \dots, K\}$) represents the controller for the k th model in the i th sub-model, $e_{ik} = r_i - \hat{w}_{ik}$ is the tracking error and μ_{ik} is obtained from the stability analysis. We utilize the following assumptions and lemmas to design a proper controller.

Assumption 1. We assume that the un-modeled subsystem's states have bounded values. It means all signals are bounded and dynamics can be modeled.

Lemma 1. Consider a function as $v^2(t)$, then fractional derivative of order p of $v^2(t)$ complies the subsequent inequality,

$$C_t^p v^2(t) \leq v(t) C_t^p v(t), \quad \forall p \in (0, 1) \quad (19)$$

Lemma 2. Suppose that Λ_{11} , Λ_{12} and Λ_{22} are the symmetric positive matrices. There are inequalities $\Lambda_{11} - \Lambda_{12}^T \Lambda_{22}^{-1} \Lambda_{12} > 0$ and $\Lambda_{22} - \Lambda_{12}^T \Lambda_{11}^{-1} \Lambda_{12} > 0$ in a such way that

$$\begin{bmatrix} \Lambda_{11} & \Lambda_{12} \\ \Lambda_{12} & \Lambda_{22} \end{bmatrix} < 0 \quad (20)$$

Lemma 3. Let Ξ , e , and η be real matrices such that

$$\Xi^T(t)\Xi(t) \leq I, \quad \eta\Xi(t)e + e^T\Xi(t)\eta^T < 0 \quad (21)$$

Then, by defining a positive scalar ρ , we deduce the following inequality.

$$\rho\eta\eta^T + \rho^{-1}\Xi^T(t)\Xi(t) < 0 \quad (22)$$

Lemma 4. The system $C_t^p v(t) = Av(t)$, $0 < p < 1$ is asymptotically stable, if

$$\sum_{i=1}^2 \sum_{j=1}^2 \text{Sym}\{\Gamma_{ij} \otimes (AQ_{ij})\} < 0, \quad (23)$$

$$\begin{bmatrix} Q_{11} & Q_{12} \\ -Q_{12} & Q_{11} \end{bmatrix} > 0, \quad \begin{bmatrix} Q_{21} & Q_{22} \\ -Q_{22} & Q_{21} \end{bmatrix} > 0$$

where $\otimes$ is the Kronecker product, Q_{11} and Q_{21} are symmetric matrices, Q_{12} and Q_{22} are skew symmetric matrices (see Ghaderpour (2016) for definitions), $\text{Sym}\{A\}$ is described as $\text{Sym}\{A\} = A + A^T$, and Γ_{ij} ($i, j \in \{1, 2\}$) are as follows.

$$\begin{aligned} \Gamma_{11} &= \begin{bmatrix} \sin(p\pi/2) & -\cos(p\pi/2) \\ \cos(p\pi/2) & \sin(p\pi/2) \end{bmatrix} \\ \Gamma_{12} &= \begin{bmatrix} \cos(p\pi/2) & \sin(p\pi/2) \\ -\sin(p\pi/2) & \cos(p\pi/2) \end{bmatrix} \\ \Gamma_{21} &= \begin{bmatrix} \sin(p\pi/2) & \cos(p\pi/2) \\ -\cos(p\pi/2) & \sin(p\pi/2) \end{bmatrix} \\ \Gamma_{22} &= \begin{bmatrix} -\cos(p\pi/2) & \sin(p\pi/2) \\ -\sin(p\pi/2) & -\cos(p\pi/2) \end{bmatrix} \end{aligned} \quad (24)$$

7. Stability analysis

Theorem 1. Consider the i th sub-model for MIPS. If there exist adaption laws such that

$$C^{p_{ik}} \phi_{ik} = -\lambda \hat{e}_{ik} \xi, \quad (25)$$

then the i th sub-model is stable in the sense of asymptotically. Parameter λ is adaption rate and there exist Φ_{ik} , χ_{ik} , τ_{ik} ($k \in \{1, \dots, K\}$) and real scalars ρ_j , $j \in \{1, 2\}$ such that

$$\begin{bmatrix} \Lambda_{11} & \Lambda_{12} \\ \Lambda_{12} & \Lambda_{22} \end{bmatrix} < 0 \quad (26)$$

where Λ_{ij} , ($i, j \in \{1, 2\}$) are defined as:

$$\begin{aligned} \Lambda_{11} &= \sum_{j=1}^2 \text{Sym} \left\{ \Gamma_{j1} \otimes \begin{bmatrix} -\tau_{i1} & 0 & \xi^T & \dots & 0 & 0 & \underline{0} \\ -\tau_{i1} & -\chi_{i1} & \underline{0} & \dots & 0 & 0 & \underline{0} \\ -\lambda \xi & \underline{0} & \underline{0} & \dots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \dots & -\tau_{iK} & 0 & \xi^T \\ 0 & 0 & \underline{0} & \dots & -\tau_{iK} & -\chi_{iK} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \dots & -\lambda \xi & \underline{0} & \underline{0} \end{bmatrix} \right\} \\ &+ \sum_{j=1}^2 \rho_j \left\{ I_2 \otimes \begin{bmatrix} 0 & \eta_{i1} \eta_{i1}^T & \underline{0} & \dots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \dots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \dots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \dots & 0 & \eta_{iK} \eta_{iK}^T & \underline{0} \\ 0 & 0 & \underline{0} & \dots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \dots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \right\} \end{aligned} \quad (27)$$

$$\Lambda_{22} = [I_2 \otimes \Phi^T \cdot I_2 \otimes \Phi^T] \quad (28)$$

where

$$\begin{aligned} \Phi &= \begin{bmatrix} 0 & 0 & \underline{0} & \dots & 0 & 0 & \underline{0} \\ 0 & \Phi_{i1} & \underline{0} & \dots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \dots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \dots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \dots & 0 & \Phi_{iK} & \underline{0} \\ 0 & \underline{0} & \underline{0} & \dots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \\ \Lambda_{12} &= \begin{bmatrix} -\rho_1 & 0 \\ 0 & -\rho_1 \end{bmatrix} \otimes I_2 \end{aligned} \quad (29)$$

Remark 3. The symbol $\underline{0}$ describes a scalar, and $\underline{0}$ determines a vector with proper dimension and zero entries.

Proof. By substituting Eq. (18) into system (16), the dynamics of tracking error can be written as:

$$\begin{cases} C_t^{p_{11}} e_{11}(t) = -\mu_{11} e_{11} - \tau_{11} \hat{e}_{11} \\ \vdots \\ C_t^{p_{1K}} e_{1K}(t) = -\mu_{1K} e_{1K} - \tau_{1K} \hat{e}_{1K} \\ \vdots \\ C_t^{p_{n1}} e_{n1}(t) = -\mu_{n1} e_{n1} - \tau_{n1} \hat{e}_{n1} \\ \vdots \\ C_t^{p_{nK}} e_{nK}(t) = -\mu_{nK} e_{nK} - \tau_{nK} \hat{e}_{nK} \end{cases} \quad (30)$$

From system (16) and by considering optimal fuzzy systems $\hat{F}_{ik}^*$ [$w(t)|\phi_{ik}^*(t)$], the estimation errors dynamics can be described by

$$\begin{cases} C_t^{p11} \hat{e}_{11}(t) = \hat{F}_{11}[w(t)|\phi_{11}(t)] - \hat{F}_{11}^*[w(t)|\phi_{11}^*(t)] + C_t^{p11} w_1(t) \\ \quad - C_t^{p11} w_1(t) - \tau_{11} \hat{e}_{11} \\ \quad \vdots \\ C_t^{p1K} \hat{e}_{1K}(t) = \hat{F}_{1K}[w(t)|\phi_{1K}(t)] - \hat{F}_{1K}^*[w(t)|\phi_{1K}^*(t)] + C_t^{p11} w_1(t) \\ \quad - C_t^{p1K} w_1(t) - \tau_{1K} \hat{e}_{1K} \\ \quad \vdots \end{cases} \quad (31)$$

$$\begin{cases} C_t^{pn1} \hat{e}_{n1}(t) = \hat{F}_{n1}[w(t)|\phi_{n1}(t)] - \hat{F}_{n1}^*[w(t)|\phi_{n1}^*(t)] + C_t^{pn1} w_n(t) \\ \quad - C_t^{pn1} w_n(t) - \tau_{n1} \hat{e}_{n1} \\ \quad \vdots \\ C_t^{pnK} \hat{e}_{nK}(t) = \hat{F}_{nK}[w(t)|\phi_{nK}(t)] - \hat{F}_{nK}^*[w(t)|\phi_{nK}^*(t)] + C_t^{pn1} w_n(t) \\ \quad - C_t^{pnK} w_n(t) - \tau_{nK} \hat{e}_{nK} \end{cases}$$

The output of $\hat{F}_{ik}[w(t)|\phi_{ik}(t)]$ is obtained as follows:

$$\hat{F}_{ik} = \phi_{ik} \xi \quad (32)$$

It follows from Eq. (32) that:

$$\begin{aligned} \hat{F}_{ik}[w(t)|\phi_{ik}(t)] - \hat{F}_{ik}^*[w(t)|\phi_{ik}^*(t)] &= \tilde{\phi}_{ik}^T \xi, \quad i \in \{1, 2, \dots, n\}, \\ k &\in \{1, 2, \dots, K\}. \end{aligned} \quad (33)$$

The approximation error ε_{ik} is given by:

$$\varepsilon_{ik} = C_t^{pi} w_i(t) - C_t^{qik} w_i(t), \quad i \in \{1, 2, \dots, n\}, \quad k \in \{1, 2, \dots, K\}. \quad (34)$$

Assumption 2. There exists constant η_{ik} , such that

$$\varepsilon_{ik} = \eta_{ik} \Xi_{ik}(t) e_{ik}, \quad |\Xi_{ik}(t)| \leq 1 \quad (35)$$

Because of the impact of the estimation errors on the tracking error, this is a logical assumption.

According to Eqs. (33) and (34), the dynamics of the estimation errors can be expressed as:

$$\begin{cases} C_t^{p11} \hat{e}_{11} = \tilde{\phi}_{11}^T \xi + \varepsilon_{11} - \tau_{11} \hat{e}_{11} \\ \quad \vdots \\ C_t^{p1K} \hat{e}_{1K} = \tilde{\phi}_{1K}^T \xi + \varepsilon_{1K} - \tau_{1K} \hat{e}_{1K} \\ \quad \vdots \\ C_t^{pn1} \hat{e}_{n1} = \tilde{\phi}_{n1}^T \xi + \varepsilon_{n1} - \tau_{n1} \hat{e}_{n1} \\ \quad \vdots \\ C_t^{pnK} \hat{e}_{nK} = \tilde{\phi}_{nK}^T \xi + \varepsilon_{nK} - \tau_{nK} \hat{e}_{nK} \end{cases} \quad (36)$$

Considering the i th subsystem and its estimated models, the matrix form can be written as follows:

$$\begin{bmatrix} C_t^{p11} \hat{e}_{i1} \\ C_t^{p11} e_{i1} \\ C_t^{p11} \tilde{\phi}_{i1} \\ \vdots \\ C_t^{p1K} \hat{e}_{iK} \\ C_t^{p1K} e_{iK} \\ C_t^{p1K} \tilde{\phi}_{iK} \end{bmatrix} = \begin{bmatrix} -\tau_{i1} & \eta_{i1} \Xi_{i1}(t) & \xi^T & \cdots & 0 & 0 & \underline{0}_{1 \times L} \\ -\tau_{i1} & -\mu_{i1} & \underline{0}_{1 \times L} & \cdots & 0 & 0 & \underline{0}_{1 \times L} \\ -\lambda \xi & \underline{0}_{L \times 1} & \underline{0}_{L \times L} & \cdots & \underline{0}_{L \times 1} & \underline{0}_{L \times 1} & \underline{0}_{L \times L} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0}_{1 \times L} & \cdots & -\tau_{iK} & \eta_{iK} \Xi_{iK}(t) & \xi^T \\ 0 & 0 & \underline{0}_{1 \times L} & \cdots & -\tau_{iK} & -\mu_{iK} & \underline{0}_{1 \times L} \\ \underline{0}_{L \times 1} & \underline{0}_{L \times 1} & \underline{0}_{L \times L} & \cdots & -\lambda \xi & \underline{0}_{L \times 1} & \underline{0}_{L \times L} \end{bmatrix} \times \begin{bmatrix} \hat{e}_{i1} \\ e_{i1} \\ \tilde{\phi}_{i1} \\ \vdots \\ \hat{e}_{iK} \\ e_{iK} \\ \tilde{\phi}_{iK} \end{bmatrix} \quad (37)$$

In the light of Lemma 4 and according to:

$$Q_{11} = Q_{21} = \begin{bmatrix} 1 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & \Phi_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & 1 & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 1 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & \Phi_{iK} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & 1 \end{bmatrix} \quad (38)$$

system (37) is asymptotically stable if:

$$\sum_{j=1}^2 \text{Sym} \left\{ \Gamma_{j1} \otimes \begin{bmatrix} -\tau_{i1} & \eta_{i1} \Xi_{i1}(t) & \xi^T & \cdots & 0 & 0 & \underline{0} \\ -\tau_{i1} & -\mu_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ -\lambda \xi & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & -\tau_{iK} & \eta_{iK} \Xi_{iK}(t) & \xi^T \\ 0 & 0 & \underline{0} & \cdots & -\tau_{iK} & -\mu_{iK} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & -\lambda \xi & \underline{0} & \underline{0} \end{bmatrix} \right\} < 0 \quad (39)$$

where $\underline{0}$ is zero vector with appropriate dimension. Inequality (18) can be described by:

$$\begin{aligned} & \sum_{j=1}^2 \text{Sym} \left\{ \Gamma_{j1} \otimes \begin{bmatrix} -\tau_{i1} & 0 & \xi^T & \cdots & 0 & 0 & \underline{0} \\ -\tau_{i1} & -\mu_{i1} \Phi_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ -\lambda \xi & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & -\tau_{iK} & 0 & \xi^T \\ 0 & 0 & \underline{0} & \cdots & -\tau_{iK} & -\mu_{iK} \Phi_{iK} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & -\lambda \xi & \underline{0} & \underline{0} \end{bmatrix} \right\} \\ & + \sum_{j=1}^2 \text{Sym} \left\{ \Gamma_{j1} \otimes \begin{bmatrix} 0 & \eta_{i1} \Xi_{i1}(t) \Phi_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 0 & \eta_{iK} \Xi_{iK}(t) \Phi_{iK} & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \right\} < 0 \end{aligned} \quad (40)$$

The second term in inequality (40) is equivalent to

$$\text{Sym} \{ \Gamma_{j1} \otimes (\Psi \cdot \Xi \cdot \Phi) \} = \text{Sym} \{ (\Gamma_{j1} \otimes \Psi) \cdot (I_2 \otimes \Xi) \cdot (I_2 \otimes \Phi) \} \quad (41)$$

where

$$\Psi = \begin{bmatrix} 0 & \eta_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 0 & \eta_{iK} & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \quad \text{and}$$

$$\Xi = \begin{bmatrix} 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & \Xi_{i1}(t) & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & \Xi_{iK}(t) & \underline{0} \\ 0 & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix}$$

We have

$$\begin{aligned} & (I_2 \otimes \Xi) \cdot (I_2 \otimes \Xi)^T \\ &= I_2 \otimes \begin{bmatrix} 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & \Xi_{i1}^T(t)\Xi_{i1}(t) & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & \Xi_{iK}^T(t)\Xi_{iK}(t) & \underline{0} \\ 0 & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \leq I \end{aligned} \quad (42)$$

Based on Lemma 3 and using Eqs. (41) and (42), we obtain

$$\begin{aligned} & \text{Sym} \left\{ \Gamma_{j1} \otimes \begin{bmatrix} 0 & \eta_{i1}\Xi_{i1}(t)\Phi_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 0 & \eta_{iK}\Xi_{iK}(t)\Phi_{iK} & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \right\} \\ & \leq \rho_j (\Gamma_{j1} \otimes \Psi) \cdot (\Gamma_{j1} \otimes \Psi)^T + \rho_j^{-1} (I_2 \otimes \Phi)^T \cdot (I_2 \otimes \Phi) \end{aligned} \quad (43)$$

From Eq. (43) and applying the fact that $\Gamma_{j1} \otimes \Gamma_{j1}^T$ ($j \in \{1, 2\}$), inequality (40), defines as:

$$\begin{aligned} & \sum_{j=1}^2 \text{Sym} \left\{ \Gamma_{j1} \otimes \begin{bmatrix} -\tau_{i1} & 0 & \xi^T & \cdots & 0 & 0 & \underline{0} \\ -\tau_{i1} & -\mu_{i1}\Phi_{i1} & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ -\lambda\xi & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & -\tau_{iK} & 0 & \xi^T \\ 0 & 0 & \underline{0} & \cdots & -\tau_{iK} & -\mu_{iK}\Phi_{i1} & \underline{0} \\ 0 & \underline{0} & \underline{0} & \cdots & -\lambda\xi & \underline{0} & \underline{0} \end{bmatrix} \right\} \\ & + \sum_{j=1}^2 \left\{ \rho_j I_2 \otimes \begin{bmatrix} 0 & \eta_{i1}\eta_{i1}^T & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \underline{0} & \cdots & 0 & \eta_{iK}\eta_{iK}^T & \underline{0} \\ 0 & 0 & \underline{0} & \cdots & 0 & 0 & \underline{0} \\ 0 & \underline{0} & \underline{0} & \cdots & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} \right. \\ & \left. + \rho_j^{-1} (I_2 \otimes \Phi)^T \cdot (I_2 \otimes \Phi) \right\} < 0 \end{aligned} \quad (44)$$

The second term in inequality (44) can be reformulated as:

$$\begin{aligned} & \sum_{j=1}^2 \rho_j^{-1} (I_2 \otimes \Phi)^T \cdot (I_2 \otimes \Phi) \\ &= [I_2 \otimes \Phi^T \quad I_2 \otimes \Phi^T] \cdot \left\{ \begin{bmatrix} \rho_1 & 0 \\ 0 & \rho_2 \end{bmatrix} \otimes I_2 \right\}^{-1} \cdot [I_2 \otimes \Phi^T \quad I_2 \otimes \Phi^T]^T \end{aligned} \quad (45)$$

By setting $\chi_{ik} = \mu_{ik}\Phi_{ik}$ ($k \in \{1, 2, \dots, K\}$) and using (44), (45) and Lemma 2, inequality (26) can be obtained. This concludes the proof. For the future studies, the robust controls and hybrid optimization techniques can be used as mentioned in Tsai and Wang (2002), Yebi and Ayalew (2016) and Zhang et al. (2023).

8. Simulations

In this paper, scenario reduction with the Monte Carlo method is employed for simulating wind energy. To represent the variations of wind farms, $N_{ss} = 100$ random initial scenarios are generated by Monte Carlo sampling. Then, the scenario reduction method is used to obtain $N_s = 10$ representative scenarios. Wind energy changes are shown in Fig. 5.

The proposed scheme is examined on a IEEE 33-bus test system (3715 kW/2300 kVar). The test system in areas 1–3 is divided, the generators G_1 , G_4 , and G_9 are responsible for LFC, respectively. To examine the accuracy of the controller, a time-variant parameter is added to the system. The delay time of 0.2 s and coordination factor of 0.3 is considered. Governor dead-band (GDB), GRC, capacity limitation of BESS, and wind speed limit are considered. Type-2 Fuzzy method is designed with the purpose of frequency adjustment. The Gaussian MFs are considered and the error MMFs are $-0.7, 0$, and 0.7 , respectively. MFs (S), (M), and (H) are named small, medium, and large, respectively. Also, Input MMFs are named “negative” (N), “zero” (Z) and “positive” (P). To evaluate the proposed method, the classical controller (PI) and type-2 fuzzy controller are optimized. The best gains can be obtained as -0.55 and -0.5 by using the K_p and K_i controller coefficients, respectively. In this study, different scenarios have been investigated considering DR, BESS, and wind farms.

As shown in the simulation results, the designed method has a good condition in terms of transient response and steady state. It has reached a steady state with a smaller fluctuation range than other methods. The type-2 fuzzy control has smaller fluctuating changes than the PI (see Table 1 for numerical comparison). On the other hand, in the introduced method, the fluctuating changes have reached a steady state in $t = 40$ s. In the following, the results of changes in the frequency of area 1 have been examined by considering DR and BESS. From Fig. 6(a), despite the DR and BESS, the frequency changes of the proposed method have a more favorable situation. Fig. 6(b)–(d) show the frequency changes of areas 1, 2 and 3 despite all scenarios. As can be seen, it immediately reaches a steady state after the disturbance. Also, the low-frequency variations of the output response indicate wind farms in this scenario. In the proposed controller, Overshoot is around 0.01, which is the minimum value. Based on the output response results, the proposed controller exhibits the best performance and the PI controller is the worst-performing control method.

To show the effectiveness of the schemed controller, Fig. 7(a)–(c) explain the comparison of frequency deviations of the conventional PI, type-2 FLC, and proposed controller, respectively. With the addition of DR, the fluctuations of frequency divisions are reduced because part of the load changes are supplied by consumers. Based on the simulation results, by adding BESS, the frequency changes are reduced, but the wind power creates small fluctuations and does not have a significant effect on the amplitude of the frequency changes.

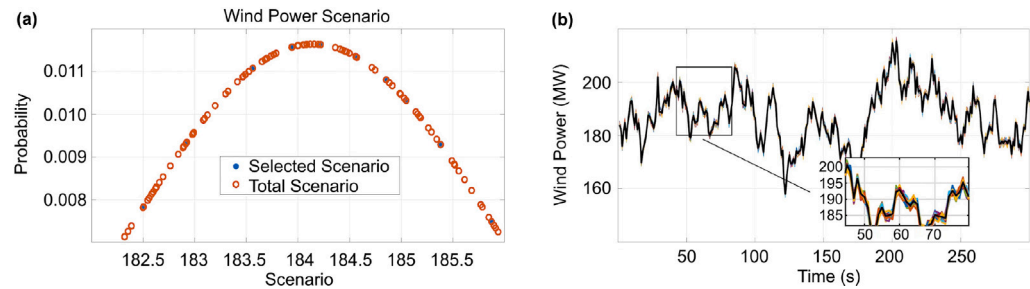


Fig. 5. (a) Scenario reduction method generated by Monte Carlo ($N_{ss} = 100$, $N_s = 10$), and (b) Total wind power generation.

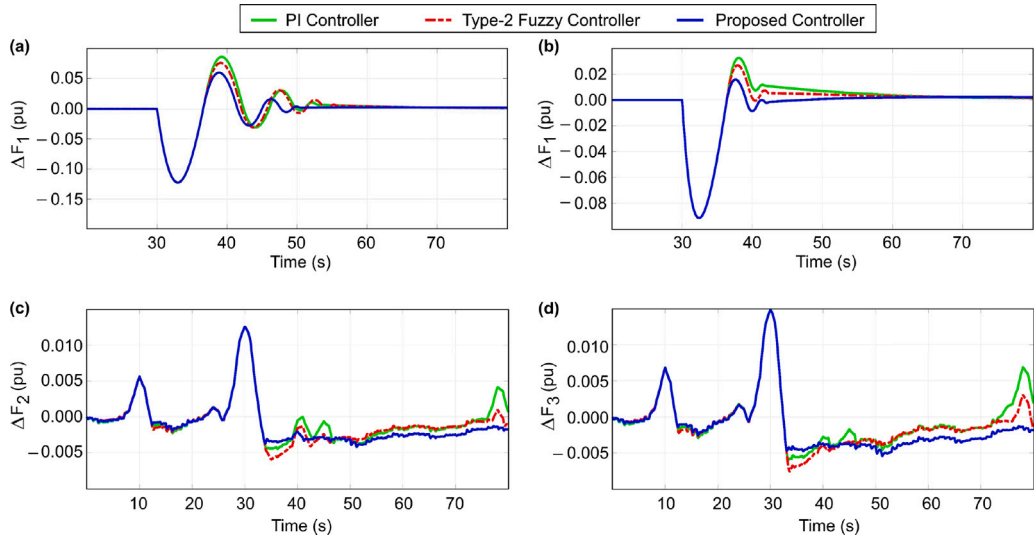


Fig. 6. Frequency deviation in area 1 considering DR and BESS, (b)–(d) Comparison of frequency deviation in areas 1, 2 and 3, respectively.

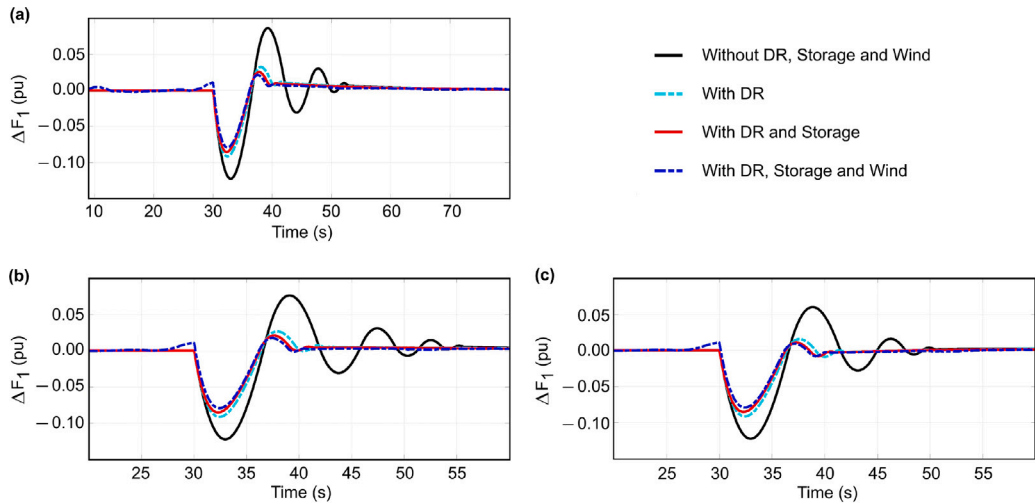


Fig. 7. Frequency deviation in area 1 with (a) PI controller, (b) Type 2 fuzzy controller, and (c) with proposed controller.

Table 1
The mean squared error (MSE) comparison between the models.

Area	PI controller (Sahu et al., 2014)	Type-2 fuzzy controller (Shakibjoo et al., 2021)	Proposed method
1	0.9147	0.8142	0.6127
2	0.8047	0.7880	0.5017
3	0.8741	0.8107	0.6112

9. Conclusion

This paper presents a novel approach to modeling and controlling load frequency in multi-region power systems using multiple first-order dynamic fractional-order T2-FLSs. The models are evaluated to identify the most effective dynamic model, from which an optimal controller is designed. The stability and optimality of the system are analyzed using a LMI approach. Furthermore, adaptive laws for the T2-FLSs are developed to ensure that specific LMI-based conditions are met. To validate the proposed approach, the 39-bus IEEE test system, incorporating wind farms, demand response, and energy storage systems, is utilized. Through extensive simulation under diverse conditions, the efficacy of the suggested controller is confirmed. The results demonstrate the practical utility of the proposed method in effectively controlling complex dynamic systems. Also, the proposed method is compared with the conventional controllers in various scenarios, and the results demonstrate the superiority of the suggested scheme. For the future studies, it is recommended new optimization techniques and FLSs to be used to improve the modeling accuracy. The effect of the challenges such as time-delay can also be considered.

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CRedit authorship contribution statement

Shu-Rong Yan: Writing – original draft, Visualization, Conceptualization. **Ying Dai:** Writing – review & editing, Conceptualization. **Ali Dokht Shakibjoo:** Writing – review & editing, Conceptualization. **Lixing Zhu:** Writing – review & editing, Conceptualization. **Sima Taghizadeh:** Writing – review & editing, Conceptualization. **Ebrahim Ghaderpour:** Writing – review & editing, Visualization, Conceptualization. **Ardashir Mohammadzadeh:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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